

Additional File for Review — Not for Publication

Geodesic Diagnostic Trajectories (GDT)

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This file contains reviewer-facing technical details that support assessment of reproducibility and methodological soundness. It is not intended for publication.

Contents

1 Computational Environment

- **Hardware:** MacBook Pro (Apple M-series), 36 GB unified memory
- **OS:** macOS 15.x
- **Python:** 3.12.4 (pyenv)
- **Key libraries:** PyTorch 2.x, HuggingFace Transformers, `rank_bm25`, `scikit-learn`, `scipy`, `geomstats` (SPD manifold operations)
- **ClinicalBERT:** `emilyalsentzer/Bio_ClinicalBERT` from HuggingFace Hub
- **Random seeds:** All experiments seeded with 42 for reproducibility

2 Data Use Agreements

All datasets are publicly available under standard data use agreements. The authors hold active agreements for MIMIC-III, MIMIC-IV, eICU, and HiRID via PhysioNet. No re-identification of patient data was performed; de-identified data were used exclusively. Synthetic vignette components generated from aggregate distributional statistics do not constitute protected health information under HIPAA Safe Harbor criteria.

3 Vignette Construction Protocol

Each of the 31 test vignettes was constructed by the following procedure:

1. Select a hospitalization record from MIMIC-III/IV, eICU, or HiRID with ≥ 3 distinct diagnostic category transitions documented in progress notes or orders.
2. Identify 3–7 “pivot points” in the clinical record where the treating team’s documented focus shifted to a new problem.
3. For each pivot point, construct a natural-language query representing the information need at that moment (e.g., “current creatinine trend and eGFR trajectory”).
4. Define a gold-standard target document (the note or lab report that best answers the query) from the de-identified record.
5. Construct a retrieval corpus of 500–2000 documents from the same hospitalization plus background documents sampled from the full dataset.

Two clinical informaticists reviewed all 31 vignettes for clinical plausibility and query naturalness. Inter-rater agreement on gold-standard document selection was $\kappa = 0.81$ (substantial agreement).

4 NASA-TLX Simulation Methodology

Because no human clinicians participated, NASA-TLX subscale scores were estimated from observable system events using the following mapping:

These are explicitly labeled as *proxy estimates*, not clinician self-report, throughout the manuscript. We use the term “simulated NASA-TLX” consistently to prevent misinterpretation.

Table 1: NASA-TLX subscale proxy mapping used in simulation.

Subscale	Observable proxy	Weight
Mental Demand	Mean query complexity (token count)	20%
Temporal Demand	Mean latency / 300s budget	20%
Effort	Query reformulation rate	20%
Frustration	Failed retrievals / total queries	20%
Performance	Inverse of time-to-correct-info	20%

5 BM25 and TF-IDF Implementation Details

Listing 1: BM25 session context implementation

```

from rank_bm25 import BM25Okapi
import numpy as np

class SessionBM25:
    def __init__(self, k1=1.5, b=0.75, window=10, decay=0.9):
        self.k1 = k1; self.b = b
        self.window = window; self.decay = decay

    def build_query(self, history):
        # Recency-weighted query: concatenate last 'window' queries
        # with exponential downweighting of older tokens
        weighted = []
        for i, q in enumerate(history[-self.window:]):
            weight = self.decay ** (len(history[-self.window:]) - i - 1)
            weighted.extend(q.split() * max(1, int(weight * 10)))
        return weighted

    def retrieve(self, corpus, history, top_k=10):
        bm25 = BM25Okapi(corpus)
        query = self.build_query(history)
        return bm25.get_top_n(query, corpus, n=top_k)

```

6 Code Repository Structure

Upon acceptance, the following repository structure will be made public via GitHub:

```

gdt-clinical-retrieval/
  src/
    models/
      gdt.py          # GDT retriever (SPD + GSI gate +
JEPA)                # CMA comparison system
      cma.py          # Session-aware BM25 baseline
      bm25_session.py
      tfidf_session.py
    geometry/
      spd_manifold.py # SPD covariance, geodesic distance
      logeuclidean.py # Log-Euclidean projection
    jepa/

```

```

        predictor.py      # Latent intent predictor
        confidence.py     # Confidence scaling
    evaluation/
        crossover.py      # Four-arm crossover trial logic
        metrics.py        # Latency, NASA-TLX proxy, Prop.1
check
        queueing.py       # M/M/c model
    data/
        mimic_loader.py
        eicu_loader.py
        vignette_builder.py
    experiments/
        run_crossover.py   # Main experiment entry point
        run_ablation.py
    configs/
        gdt_default.yaml   # All hyperparameters (Table S1)
    requirements.txt
    README.md

```